

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
LABORATORY OBSERVATION & RECORD

Course Name: Full Stack Web Development

Course Code: 23CM4121

Experiment No: 04

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Branch & Section: CSM-B

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1. TITLE

Develop Dynamic Webpage Using DOM & JS

2. AIM

To design and build an interactive web page that dynamically processes form input data, modifies the DOM content in real time, and handles background style changes using JavaScript event listeners.

3. OBJECTIVE

- To link external CSS stylesheets and JavaScript files using DOMContentLoaded listeners.
- To capture form submit events, prevent default reloads, and parse data using FormData.
- To dynamically update DOM text content and modify element inline background styles.

4. THEORY

The Document Object Model (DOM) presents an HTML page as a node tree that JavaScript can query and modify dynamically. Using event listeners like DOMContentLoaded ensures DOM elements exist before binding listeners. The FormData API extracts submitted form entries efficiently, allowing client-side JS code to display form values dynamically in paragraph tags without triggering full web page reloads.

5. ALGORITHM / PROCEDURE

1. Create index.html, styles.css, and script.js files in the project workspace.
2. Construct an HTML page containing a personal information form, action button, and an output display container.
3. Apply CSS styling to form components, set up gradient background styles, and adjust box boundaries.
4. Add a DOMContentLoaded event listener in script.js to safely query and attach event listeners.
5. Attach a click listener to the background toggle button to switch background colors dynamically.

6. Intercept form submit events using `event.preventDefault()`, construct a `FormData` object, and update DOM output paragraphs with user details.
7. Open `index.html` in a web browser, test form submissions, and verify dynamic output rendering.

6. PROGRAM

```
<!-- index.html -->
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Dynamic Webpage</title>
  <link rel="stylesheet" href="./styles.css">
  <script src="./script.js"></script>
</head>
<body>
  <h1>This is a Dynamic webpage</h1>
  <form action="#submit" id="form">
    <legend>Personal Information</legend>
    <label for="fullname">Name: </label>
    <input id="fullname" name="fullname" type="text" placeholder="enter your
name"><br>
    <label for="email">Mail-Id: </label>
    <input id="email" name="email" type="email"
placeholder="email@example.com"><br>
    <label for="phone">Phone: </label>
    <input id="phone" name="phone" type="tel" min="10"
placeholder="9123456780"><br>
    <label for="address">Address: </label>
    <textarea id="address" name="address" placeholder="Enter your address"
rows="10" cols="50"></textarea>
    <button id="submit">submit</button>
  </form>
  <button type="button" id="bg">click the button to change text</button>

  <div id="output">
    <p id="output-name"></p>
    <p id="output-email"></p>
    <p id="output-phone"></p>
    <p id="output-address"></p>
  </div>
</body>
</html>

/* styles.css */
body {
  background: linear-gradient(to right, #ff7e5f, #feb47b);
  align-items: center;
}
#form {
  color: #fff;
  align-items: center;
  border: 2px solid #fff;
  border-radius: 10px;
  border-width: 2px;
  width: 50%;
  height: 50%;
}
#form label {
```

```

        font-size: 1.5rem;
        font-weight: bold;
        font-family: Arial, sans-serif;
    }
    #form legend {
        font-size: 1.5rem;
        font-weight: bold;
        font-family: Arial, sans-serif;
    }

    // script.js
    document.addEventListener("DOMContentLoaded", () => {
        const background = document.getElementById("bg");
        const form = document.getElementById("form");

        if (background) {
            background.addEventListener("click", () => {
                const nextColor = background.style.backgroundColor === "lightblue" ?
"lightgreen" : "lightblue";
                background.style.backgroundColor = nextColor;
            });
        }
        const output = document.getElementById("output");

        if (form) {
            form.addEventListener("submit", (event) => {
                event.preventDefault();

                const formData = new FormData(form);
                const submittedData = Object.fromEntries(formData.entries());

                console.log("Form data stored in variable:", submittedData);

                if (output) {
                    document.getElementById("output-name").textContent = `Name:
${submittedData.fullname}`;
                    document.getElementById("output-email").textContent = `Email:
${submittedData.email}`;
                    document.getElementById("output-phone").textContent = `Phone:
${submittedData.phone}`;
                    document.getElementById("output-address").textContent = `Address:
${submittedData.address}`;
                }
            });
        }
    });

```

7. CODE EXPLANATION

- `document.addEventListener('DOMContentLoaded')`: Ensures script executes after HTML document parsing.
- `event.preventDefault()`: Prevents the form submission from triggering a page refresh.
- `new FormData(form) & Object.fromEntries()`: Extracts form control input keys and values into an object.
- `textContent`: Updates DOM paragraph nodes with user-entered values dynamically.
- `background.style.backgroundColor`: Toggles element inline background color dynamically on click.

8. OUTPUT

Test- Case 01: Load the page in browser

Output :



The screenshot shows a webpage with an orange background. At the top left, the text "This is a Dynamic webpage" is displayed in bold black font. Below this, there is a white-bordered box containing a form titled "Personal Information" in bold orange text. The form includes four input fields: "Name:" with the placeholder "enter your name", "Mail-Id:" with the placeholder "email@example.com", "Phone:" with the placeholder "9123456780", and a larger "Address:" field with the placeholder "Enter your address". A "submit" button is located at the bottom right of the form box. Below the form box, there is a small button labeled "click the button to change text".

Test-Case 02 : Submitting empty form

Output: returns empty data with field names only

This is a Dynamic webpage

Personal Information

Name:

Mail-Id:

Phone:

Address:

Name:

Email:

Phone:

Address:

Test case 03 : submitting form with an empty field

Output :

This is a Dynamic webpage

Personal Information

Name:

Mail-Id:

Phone:

Address:

Name: surya prabhath

Email: testuser@gmail.com

Phone: 7013626242

Address:

Test Case 04 : Submitting Form with all fields

Output :

The screenshot shows a web page with an orange background. At the top, it says "This is a Dynamic webpage". Below this is a form titled "Personal Information". The form contains four input fields: "Name:" with the value "surya prabhath", "Mail-Id:" with the value "testuser@gmail.com", "Phone:" with the value "7013626242", and "Address:" with the value "vslkvsf;vmdfvbk". A "submit" button is located to the right of the address field. Below the form, there is a button labeled "click the button to change text". At the bottom, the submitted data is displayed: "Name: surya prabhath", "Email: testuser@gmail.com", "Phone: 7013626242", and "Address: vslkvsf;vmdfvbk".

This is a Dynamic webpage

Personal Information

Name: surya prabhath

Mail-Id: testuser@gmail.com

Phone: 7013626242

Address: vslkvsf;vmdfvbk

submit

click the button to change text

Name: surya prabhath

Email: testuser@gmail.com

Phone: 7013626242

Address: vslkvsf;vmdfvbk

9. RESULT

A dynamic webpage handling form submissions, real-time DOM element manipulation, and event listeners was successfully created, executed, and verified.

RUBRIC EVALUATION (Faculty Use Only)

Design & Logic (5 Marks): _____

Execution & Output (5 Marks): _____

Viva & Record (5 Marks): _____

TOTAL MARKS: [/ 15]

Faculty Signature: _____